

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based Problem-Solving Question

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Industry based Problem-Solving Question	Assessment:	Assessment Tool 2 – Industry based Problem-Solving Question
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2023	Date:	18/8/2026

Code of Conduct

I N HARSHAVARDHAN(192324189) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Problem statement

An online retailer has 50,000 customers. A random sample of 500 customers shows that 68% made a purchase during a promotional campaign. Estimate the proportion of all customers who made a purchase.

1. Identify the population and sample

Population: All 50,000 customers of the online retailer. Sample: The 500 customers randomly selected during the promotional campaign.

2. Identify the parameter of interest

Population purchase proportion, p .

3. Calculate the point estimate

- $\hat{p} = x/n = 0.68$. Point estimate = 68%.

4. State the null hypothesis H_0

Assume an industry benchmark purchase rate of 65%: $H_0: p = 0.65$.

5. State the alternative hypothesis H_1

$H_1: p \neq 0.65$.

6. Select the significance level α

$\alpha = 0.05$.

7. Calculate the confidence interval and/or test statistic

- 95% CI: $\hat{p} \pm 1.96\sqrt{[\hat{p}(1 - \hat{p})/n]}$.
- $SE = \sqrt{[0.68(0.32)/500]} = 0.02087$.
- $CI = 0.68 \pm 1.96(0.02087) = (0.6391, 0.7209)$.
- Test statistic: $z = (0.68 - 0.65)/\sqrt{[0.65(0.35)/500]} = 1.406$.

8. Calculate the p-value

For a two-sided z-test, $p\text{-value} = 2P(Z > 1.406) \approx 0.1596$.

9. Compare p-value with α

$0.1596 > 0.05 \rightarrow$ Fail to reject H_0 .

10. Make an industry-oriented conclusion

The estimated purchase rate is 68%. The observed campaign conversion is not significantly different from the 65% benchmark at the 5% level. The 95% confidence interval of 63.91%–72.09% gives the retailer a practical range for the overall purchase rate.

Key takeaway: Purchase proportion = 68%; 95% CI = 63.91%–72.09%; p-value = 0.1596.

2. Problem statement

A factory produces electronic components. From a random sample of 100 components, the average lifetime is 1,250 hours. Determine the point estimate of the population mean lifetime.

1. Identify the population and sample

Population: All electronic components produced by the factory. Sample: The 100 randomly selected components whose lifetimes were measured.

2. Identify the parameter of interest

Population mean component lifetime, μ .

3. Calculate the point estimate

- $\bar{x} = 1,250$ hours. Point estimate of $\mu = 1,250$ hours.

4. State the null hypothesis H_0

Assume a manufacturing benchmark mean lifetime of 1,200 hours: $H_0: \mu = 1,200$ hours.

5. State the alternative hypothesis H_1

$H_1: \mu \neq 1,200$ hours.

6. Select the significance level α

$\alpha = 0.05$.

7. Calculate the confidence interval and/or test statistic

- Because the problem does not provide s , an illustrative $s = 200$ hours is assumed.
- $SE = s/\sqrt{n} = 200/10 = 20$ hours.
- With $df = 99$, $t_{0.975,99} \approx 1.984$.
- 95% CI = $1,250 \pm 1.984(20) = (1,210.32, 1,289.68)$ hours.
- Test statistic: $t = (1,250 - 1,200)/20 = 2.50$.

8. Calculate the p-value

For a two-sided t-test with 99 degrees of freedom, p-value ≈ 0.0141 .

9. Compare p-value with α

$0.0141 < 0.05 \rightarrow \text{Reject } H_0$.

10. Make an industry-oriented conclusion

The estimated mean component lifetime is 1,250 hours. The test indicates that the production mean is significantly different from, and in this sample higher than, the 1,200-hour benchmark. This supports the interpretation that component durability is performing above the stated benchmark.

Key takeaway: Mean lifetime = 1,250 h; 95% CI = 1,210.32–1,289.68 h; p-value = 0.0141.

3. Problem statement

A bank wants to estimate the average monthly transaction value of its customers. A sample of 200 customers has a mean transaction value of ₹18,500 and a standard deviation of ₹4,200. Calculate the point estimates of the population mean and variance.

1. Identify the population and sample

Population: All customers represented by the bank study. Sample: The 200 customers selected for measurement of monthly transaction value.

2. Identify the parameter of interest

Population mean monthly transaction value μ and population variance σ^2 .

3. Calculate the point estimate

- $\hat{\mu} = \bar{x} = ₹18,500$.
- $\sigma^2 \text{ estimate} = s^2 = (\text{₹}4,200)^2 = ₹17,640,000$.

4. State the null hypothesis H_0

Assume an industry benchmark average monthly transaction value of ₹18,000: $H_0: \mu = ₹18,000$.

5. State the alternative hypothesis H_1

$H_1: \mu \neq ₹18,000$.

6. Select the significance level α

$\alpha = 0.05$.

7. Calculate the confidence interval and/or test statistic

- $SE = 4,200/\sqrt{200} \approx ₹296.98$.
- With $df = 199$, $t_{0.975,199} \approx 1.972$.

- 95% CI = ₹18,500 ± 1.972(₹296.98) ≈ (₹17,914.36, ₹19,085.64).

- Test statistic: $t = (18,500 - 18,000)/296.98 \approx 1.684$.

8. Calculate the p-value

For a two-sided t-test with 199 degrees of freedom, $p\text{-value} \approx 0.0938$.

9. Compare p-value with α

$0.0938 > 0.05 \rightarrow \text{Fail to reject } H_0$.

10. Make an industry-oriented conclusion

The estimated average monthly transaction value is ₹18,500 and the estimated population variance is ₹17,640,000. Although the sample mean is above ₹18,000, the difference is not statistically significant at the 5% level. The bank should therefore avoid claiming a definite increase based on this sample alone.

Key takeaway: Mean = ₹18,500; variance = ₹17,640,000; 95% CI = ₹17,914.36–₹19,085.64; $p\text{-value} = 0.0938$.

4. Problem statement

A hospital randomly selects 400 patients, of whom 52 experienced a particular side effect. Find the point estimate of the proportion of patients experiencing the side effect. Explain what the estimate means.

1. Identify the population and sample

Population: All patients receiving the relevant treatment or service represented by the study. Sample: The 400 randomly selected patients.

2. Identify the parameter of interest

Population proportion p of patients experiencing the side effect.

3. Calculate the point estimate

- $\hat{p} = 52/400 = 0.13$. Point estimate = 13%.

4. State the null hypothesis H_0

Assume a clinical monitoring benchmark of 10%: $H_0: p = 0.10$.

5. State the alternative hypothesis H_1

$H_1: p \neq 0.10$.

6. Select the significance level α

$$\alpha = 0.05.$$

7. Calculate the confidence interval and/or test statistic

- 95% CI = $0.13 \pm 1.96\sqrt{[0.13(0.87)/400]} = (0.0970, 0.1630)$.
- Thus, the estimated side-effect proportion is approximately 9.70%–16.30%.
- Test statistic: $z = (0.13 - 0.10)/\sqrt{[0.10(0.90)/400]} = 2.00$.

8. Calculate the p-value

For a two-sided z-test, $p\text{-value} = 2P(Z > 2.00) \approx 0.0455$.

9. Compare p-value with α

$0.0455 < 0.05 \rightarrow \text{Reject } H_0$.

10. Make an industry-oriented conclusion

The estimated side-effect rate is 13%, meaning about 13 out of every 100 comparable patients experienced the side effect in the sample. The rate is statistically above the 10% benchmark at the 5% level, so the hospital should review safety monitoring, patient counselling, and treatment protocols.

Key takeaway: Side-effect proportion = 13%; 95% CI = 9.70%–16.30%; p-value = 0.0455.

5. Problem statement

A food-processing company measures the weight of 64 randomly selected packets. The mean weight is 498 g and the sample standard deviation is 12 g. Construct a 95% confidence interval for the true average packet weight.

1. Identify the population and sample

Population: All packets produced by the company under the relevant production process. Sample: The 64 randomly selected packets whose weights were measured.

2. Identify the parameter of interest

True population mean packet weight μ .

3. Calculate the point estimate

- $\bar{x} = 498$ g. Point estimate of $\mu = 498$ g.

4. State the null hypothesis H_0

Assume the company's target average packet weight is 500 g: $H_0: \mu = 500$ g.

5. State the alternative hypothesis H_1

$H_1: \mu \neq 500$ g.

6. Select the significance level α

$\alpha = 0.05$.

7. Calculate the confidence interval and/or test statistic

- Since σ is unknown, use the one-sample t distribution with $df = 63$.
- $SE = 12/\sqrt{64} = 1.5$ g.
- Critical value: $t_{0.975,63} \approx 1.998$.
- 95% CI = $498 \pm 1.998(1.5) = (495.00, 501.00)$ g.
- Test statistic: $t = (498 - 500)/1.5 = -1.333$.

8. Calculate the p-value

For a two-sided t-test with 63 degrees of freedom, $p\text{-value} \approx 0.1872$.

9. Compare p-value with α

$0.1872 > 0.05 \rightarrow$ Fail to reject H_0 .

10. Make an industry-oriented conclusion

The 95% confidence interval for the true average packet weight is 495.00 g to 501.00 g. Because the 500 g target lies inside this interval and the p-value exceeds 0.05, the production process is statistically consistent with the stated target average.

Key takeaway: Mean = 498 g; 95% CI = 495.00–501.00 g; p-value = 0.1872.